

VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi

Gokulam, 3rd Stage, Mysuru 570 002

Sixth Semester B.E. Examinations

Course Code: BIS/CSBT603

Model SEE Question Paper

Academic Year 2024-25

COURSE NAME: BLOCKCHAIN TECHNOLOGY

Duration: 3 hours

Max. Marks: 100

INSTRUCTION TO STUDENTS

- 1) Answer One Full question from each module
- 2) Three module questions are compulsory and the remaining two module will have internal choice

Q. No.	Module-I	Marks	BL	CO	PO
1. (a)	Explain the concept of consensus in distributed system. Classify various consensus mechanisms and the requirements to achieve consensus in distributed systems.	10	L2	CO1	PO1
1. (b)	Explain the generic elements and the tiers of blockchain technology.	10	L2	CO1	PO1
(OR)					
2.(a)	List and explain any five types of blockchain.	10	L2	CO1	PO1
2. (b)	Explain the various consensus algorithms used in blockchain.	10	L2	CO1	PO1
Module-II					
3.(a)	Explain how disintermediation removes the concept of middlemen in transaction process.	4	L2	CO1	PO1
3.(b)	Apply the concepts of full ecosystem decentralization, to achieve complete decentralization.	8	L3	CO2	PO1
3.(c)	In what ways does the autonomy of a Decentralized Autonomous Organization (DAO) impact decision-making compared to a Decentralized Organization (DO)? Apply the concept of DAOs with real-world examples to support your answer.	8	L3	CO2	PO1
Module-III					
4.(a)	With a neat diagram, apply the concept of encryption and decryption in asymmetric cryptography.	5	L3	CO2	PO1
4.(b)	Apply the process of encryption and decryption performed in RSA algorithm by considering, P=53, q=59 and message as 28.	5	L3	CO2	PO1
4.(c)	You are building a user authentication system that needs to securely store user passwords in a database. It is crucial that passwords are never stored in plaintext, and the system must be resistant to various types of attacks, such as brute-force or rainbow table attacks. Identify the suitable hashing algorithm and analyze the steps for ensuring the integrity of the password.	10	L4	CO3	PO2
Module-IV					

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5.(a)	With an example, explain OP_EQUAL and OP_DUP opcodes to verify bitcoin data.	6	L2	CO1	PO1
5.(b)	Analyze the structure of block and block header to show how the transaction is stored in blockchain.	7	L4	CO3	PO2
5.(c)	A blockchain-based marketplace uses smart contracts to automatically execute transactions between buyers and sellers. However, there is a need to ensure that the terms of these transactions are legally recognized, even outside the blockchain network, and can be enforced in a court of law if necessary. Analyze the scenario and explain the same with a neat diagram.	7	L4	CO3	PO2
Module-V					
6.(a)	“Ethereum clients (such as Geth) and the Web3.js library work together to enable the development of decentralized applications (DApps)”. Explain the key components involved in this interaction and apply the process with a neat diagram.	7	L3	CO2	PO1
6.(b)	A miner in Ethereum mines a valid block, but due to fast block generation, it becomes a stale (orphan) block. Apply how the GHOST protocol helps in determining the main chain and incentivizing such stale blocks?	7	L3	CO2	PO1
6.(C)	EVM runs bytecode instructions to transform the system from one state to another. Justify the statement.	6	L4	CO3	PO2
(OR)					
7.(a)	In the context of Ethereum, what happens when a transaction fails due to insufficient balance or gas, and how does the rollback mechanism ensure consistency in the blockchain state? Apply the concept with an example of how this process minimizes network disruptions.	7	L3	CO2	PO1
7.(b)	How does the execution environment in Ethereum leverage the block header information for transaction processing? Apply its role in maintaining blockchain integrity and consensus.	7	L3	CO2	PO1
7.(c)	How did the hard fork cause by the Homestead upgrade impact Ethereum's network and its participants? Analyze the potential risks and benefits of protocol upgrades in blockchain ecosystems.	6	L4	CO3	PO2

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